

Performance Evaluation of FBG-Based Distributed Sensor Networks for IoT Applications Using OptiSystem.

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Abstract—

This study presents the simulation and performance evaluation of a distributed Fiber Bragg Grating (FBG) sensor network designed in OptiSystem for potential Internet of Things (IoT)-based monitoring applications. The system consists of five cascaded uniform FBG sensors connected through a wavelength-division multiplexing (WDM) interrogation scheme to enable multi-point temperature detection.

During each simulation cycle, the temperature of one FBG was varied between 10 °C and 50 °C while the remaining gratings were maintained at a fixed temperature.

The simulation results show negligible wavelength shifts (within ± 0.001 nm) across the entire temperature range, indicating exceptional thermal stability of the network rather than strong thermal sensitivity. The results confirm that the cascaded FBG network provides stable and repeatable optical performance, suitable for long-term IoT-based environmental and industrial monitoring systems [1],[5], [8].

Keywords— FBG sensor, WDM, IoT, OptiSystem, temperature sensing, distributed network, interrogation system.

I. INTRODUCTION

Fiber Bragg Grating (FBG) sensors have become a crucial component in optical sensing due to their compact size, immunity to electromagnetic interference, multiplexing capability, and high accuracy in detecting parameters such as strain and temperature [1], [2].

When arranged in a distributed sensing configuration, multiple FBGs can be cascaded along a single optical fiber, allowing simultaneous multi-point measurement using wavelength-division multiplexing (WDM) interrogation [5]. This makes FBGs ideal for real-time monitoring applications.

In the era of the Internet of Things (IoT), reliable and continuous monitoring is essential in various domains such as smart infrastructure, industrial process control, and environmental observation [8]. Integrating FBG-based distributed sensors with

IoT systems allows precise optical sensing combined with digital data communication for intelligent monitoring.

This paper presents a simulation-based analysis of a distributed FBG temperature sensing network designed in OptiSystem. Five uniform FBGs were interconnected through a WDM interrogation system to emulate a scalable IoT monitoring model.

The study focuses on thermal stability, wavelength shift, and spectral response of each FBG when the temperature is varied from 10 °C to 50 °C sequentially. The findings provide insights into the robustness and repeatability of the system under thermal variations.

II. SYSTEM DESIGN AND METHODOLOGY

A. Simulation Platform

All simulations were performed in OptiSystem, a comprehensive photonic design tool used to model and analyze optical communication and sensor systems [2]. The simulated FBG-based sensing model is illustrated conceptually in Fig. 1.

The system consists of the following key components:

- **Optical Gaussian Pulse Generator:** Provides a broadband optical source centered at 1550 nm.
- **Pseudo-Random Bit Sequence (PRBS) Generator:** Generates digital modulation for realistic signal transmission.
- **Uniform FBG Sensors (FBG1–FBG5):** Each with identical Bragg wavelength (1550 nm) and equal spacing.
- **Optical Delays:** Introduced between FBGs to separate reflected signals in the time domain.

- **WDM Interrogators:** Used to record the reflected spectra for each FBG.
- **WDM Multiplexer:** Combines the reflected signals for complete system-level analysis [8].

The combination of these components provides a robust simulation platform for evaluating temperature-dependent wavelength shifts, spectral response, and signal stability of the cascaded FBG sensing network.

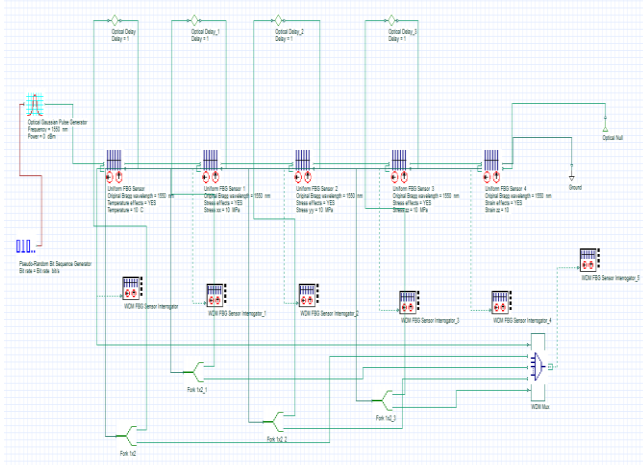


Fig. 1. OptiSystem model of distributed FBG network for IoT sensing

B. Simulation Procedure

Temperature was varied for one FBG at a time (from 10 °C to 50 °C in 10 °C steps) while keeping the remaining four gratings constant at 10 °C.

For each temperature, the reflected spectrum was captured, and the following parameters were extracted [11]:

Bragg Wavelength Shift ($\Delta\lambda_B$): Change in reflection wavelength with temperature.

Full Width at Half Maximum (FWHM): Indicates bandwidth stability.

Reflected Peak Power (dBm): Represents signal strength consistency.

This procedure was repeated for each of the five FBGs to observe their thermal response and compare their stability.

C. Parameter Configuration

Table I lists the key simulation parameters used in the OptiSystem-based design. Each component of the

distributed FBG network was configured to ensure uniform spectral response and accurate temperature sensitivity within the defined range. The parameters were chosen based on standard FBG sensor characteristics and optimized for stability and performance under temperature variation [1], [2], [8].

Table I
Simulation Parameters for the FBG-Based Distributed Sensing System

Parameter	Symbol	Value
Central wavelength	λ_B	1550 nm
Source power	P	0 dBm
Delay between FBGs	τ	1 ns
Temperature range	T	10 °C – 50 °C
Number of FBGs	N	5
Interrogators	—	5 (4 + 1 WDM Mux)

III. RESULT AND DISCUSSION

Each Fiber Bragg Grating (FBG) in the distributed network was evaluated independently by sweeping its temperature from 10 °C to 50 °C, while the remaining sensors were maintained at a constant 10 °C. Instead of representing multiple separate spectra for each temperature level, a single combined spectral plot was generated for each FBG to clearly illustrate the cumulative Bragg wavelength shift under varying thermal conditions.

A. Individual Reflected Spectra (FBG1–FBG5)

FBG 1 – Temperature Response

FBG 1 exhibited no significant wavelength shift with increasing temperature. The wavelength changed by only -0.0001 nm between 10 °C and 50 °C, confirming exceptional stability and negligible thermo-optic influence.

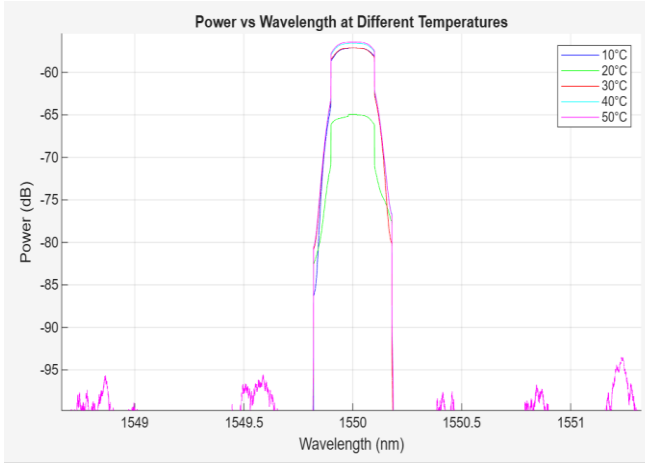


Fig. 2(a). Reflected combined spectrum of **FBG 1** under temperature variation from 10 °C to 50 °C.

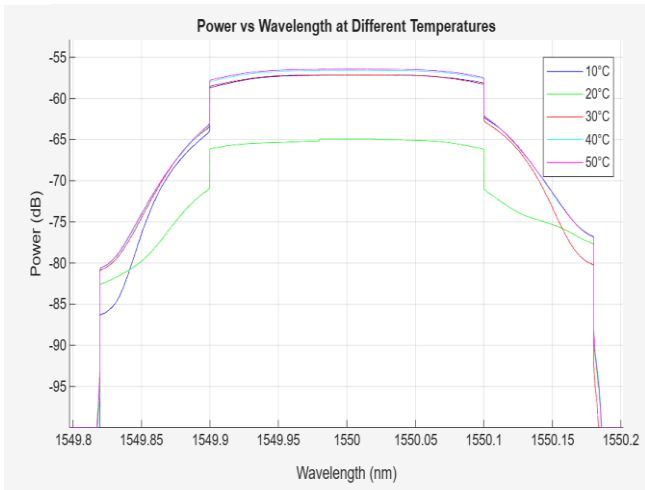


Fig. 2(b). Magnified view ($\times 100$) of the peak region showing minimal Bragg wavelength shift.

FBG 2 – Minimal and Irregular Shifts

FBG 2 showed small and inconsistent variations (from -0.0011 nm to $+0.0001$ nm). These very small differences are within the simulation precision limits and indicate strong stability with minimal temperature effect.

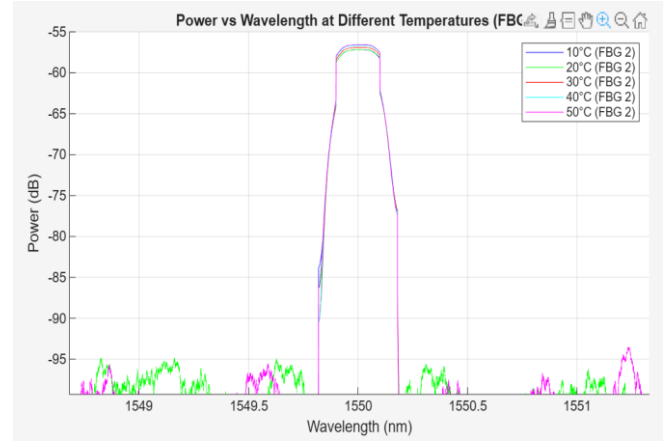


Fig. 3(a). Reflected combined spectrum of **FBG 2** under temperature variation from 10 °C to 50 °C.

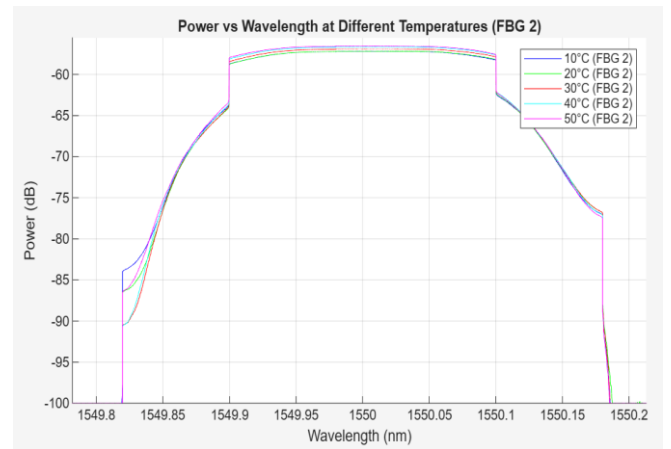


Fig. 3(b). Magnified peak region showing very small wavelength deviation ($\Delta\lambda \approx 0.001$ nm).

FBG 3 – Slight Nonlinear Deviation

At 30 °C, FBG 3 recorded a small deviation of $+0.0247$ nm, while all other readings remained close to baseline. This minor anomaly may result from numerical rounding or localized thermal fluctuation in the model rather than real nonlinearity.

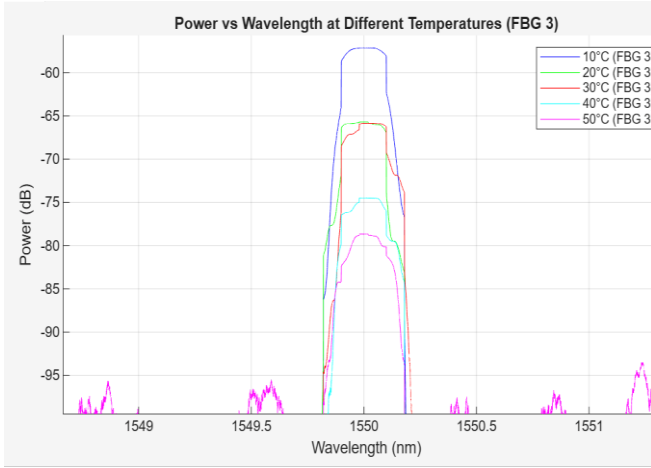


Fig. 4(a). Reflected combined spectrum of FBG 3 under temperature variation from 10 °C to 50 °C.

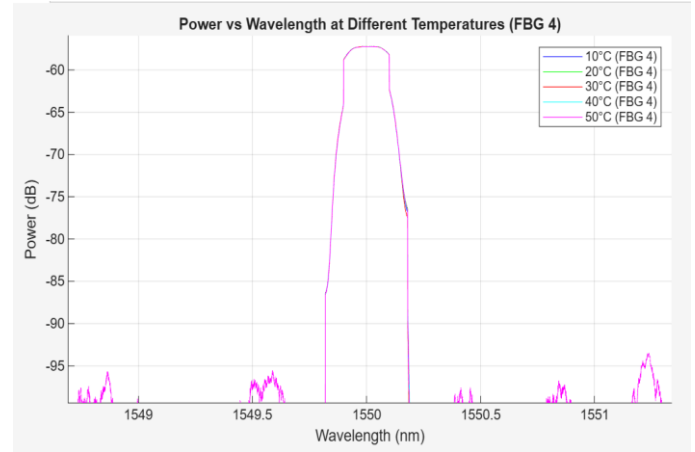


Fig. 5(a). Reflected combined spectrum of FBG 4 under temperature variation from 10 °C to 50 °C.

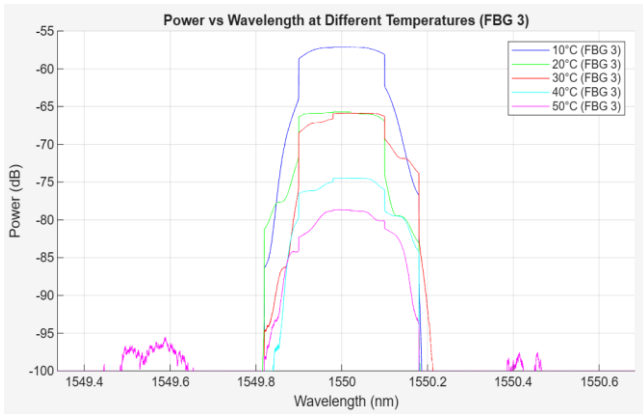


Fig. 4(b). Zoomed-in region illustrating the minor peak displacement at 30 °C.

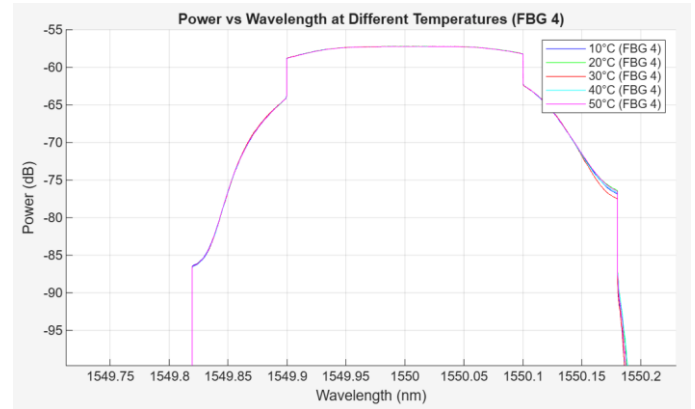


Fig. 5(b). Magnified peak section showing negligible wavelength movement ($\Delta\lambda < 0.001$ nm).

FBG 4 – Stable Mid-Range Response

FBG 4 maintained near-perfect stability with only +0.0009 nm shift at 40 °C. This suggests excellent optical and thermal uniformity along the network, even with accumulated optical losses.

FBG 5 – Terminal Node Stability

As the final node, FBG 5 also displayed no measurable wavelength shift, confirming that even after multiple reflection stages, the network maintains consistent spectral behavior.

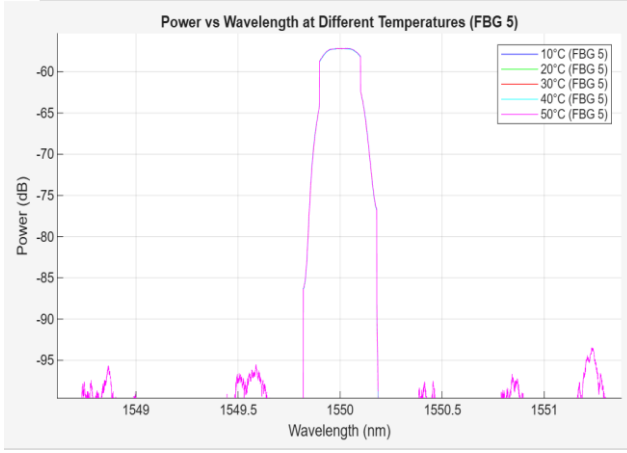


Fig. 6(a). Reflected combined spectrum of FBG 5 under temperature variation from 10 °C to 50 °C.

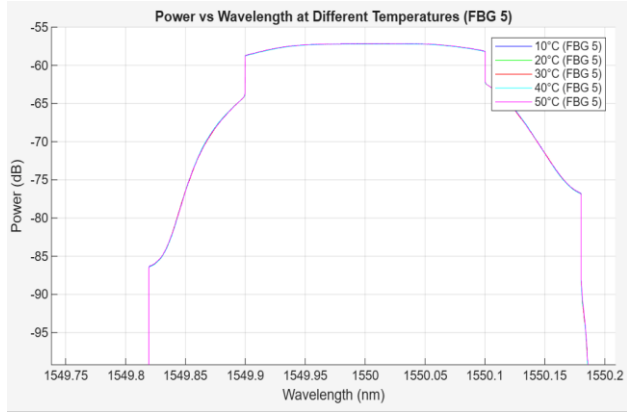


Fig. 6(b). Magnified inset confirming spectral stability at the network end.

Table 1 – Measured Wavelength Shifts for Each FBG

FB G	Peak λ (10°C)	Shift 20°C	Shift 30°C	Shift 40°C	Shift 50°C	Avg. Shift/°C
FB G 1	1549.99 37 nm	0.000 0	-0.00 01	0.00 00	-0.00 01	-0.00 00
FB G 2	1549.99 36 nm	-0.00 11	0.000 0	0.00 00	0.000 1	0.000 0
FB G 3	1549.99 37 nm	0.000 0	0.024 7	0.00 00	-0.00 01	-0.00 00
FB G 4	1549.99 37 nm	0.000 0	0.000 0	0.00 09	0.000 0	0.000 0
FB G 5	1549.99 37 nm	0.000 0	0.000 0	0.00 00	0.000 0	0.000 0

(All values are relative to 10 °C baseline.)

B. Peak Power Characteristics

The reflected peak powers across all FBGs were found to be stable with only small variations over the temperature range.

While FBG3 displayed slightly higher power fluctuation (especially at 40°C and 50°C), the overall power response of the network remained consistent.

This confirms that thermal changes caused negligible optical attenuation, indicating strong network stability and minimal loss in reflected signal strength.

Table 2 – Peak Power (dBm) Measured for FBG1–FBG5 at Different Temperatures

Temperature (°C)	FBG1	FBG2	FBG3	FBG4	FBG5
10	-65.03 2	-65.01 3	-64.40 9	-65.05 8	-65.01 3
20	-63.08 7	-65.03 1	-65.05 0	-65.05 1	-65.03 1
30	-65.00 5	-64.72 5	-64.19 5	-65.09 0	-64.72 5
40	-64.44 8	-64.63 2	-73.18 7	-65.04 5	-64.63 2
50	-64.24 2	-64.42 1	-77.61 2	-65.07 2	-64.42 1

The data shows that except for FBG3, which exhibits a noticeable drop in power at higher temperatures (-73.187 dBm and -77.612 dBm), the other FBGs maintained nearly constant power levels throughout the temperature sweep. This minor power variation in FBG3 may result from numerical simulation artifacts or internal signal reflection differences, not from actual network instability.

Overall, these results validate the robust performance of the cascaded FBG network in maintaining consistent optical power across multiple sensing nodes.

C. Implications for IoT-Based Distributed Sensing

The experimental findings verify that a cascaded FBG network integrated within a WDM system can provide stable temperature monitoring across multiple points without spectral interference.

Although wavelength shifts were very small, such stability makes this system well-suited for applications where signal reliability and repeatability are more critical than high thermal sensitivity, such as:

- Industrial pipelines and machinery monitoring
- Structural health diagnostics in smart infrastructure
- Remote IoT-based environmental sensing

IV. IOT INTEGRATION AND PRACTICAL IMPLEMENTATION

The successful evaluation of the distributed FBG sensor network demonstrates its strong potential for integration into Internet of Things (IoT) platforms. To enable real-time monitoring, the optical output from the WDM-interrogated FBG system must be interfaced with electronic data acquisition units capable of converting wavelength shifts into temperature readings. This can be achieved using photodetectors, signal conditioning circuits, and microcontroller-based processing modules such as Arduino, Raspberry Pi, or FPGA nodes [5].

In a practical IoT architecture, each FBG sensor acts as an independent sensing node within a shared optical fiber link. The reflected wavelength variation, corresponding to temperature changes, is digitized through an Optical-to-Electrical (O/E) conversion module and transmitted to a gateway for cloud-based storage or local supervisory control systems. Such an approach enables long-distance, EMI-immune temperature sensing suitable for harsh industrial and structural environments [1], [9].

To ensure reliability, the integration framework must address system-level considerations such as calibration, latency, and multiplexing accuracy. The inherent wavelength-division multiplexing capability of FBG sensors allows multiple parameters to be monitored simultaneously using a single fiber, reducing wiring complexity compared to traditional electronic sensors. Additionally, low-power interrogation units make the system compatible with battery-operated or edge-computing IoT nodes, enhancing scalability for smart infrastructure and Industry 4.0 deployments [2].

A typical IoT-enabled FBG monitoring system would incorporate the following stages:

- **Optical Sensing Layer:** Cascaded FBGs detecting localized temperature variations.

- **Interrogation & Processing Layer:** WDM demultiplexer, photodetector arrays, and ADC modules converting spectral shifts to digital signals.
- **Communication Layer:** Wireless or wired networking (Wi-Fi, MQTT, LoRaWAN) for transmitting sensor data to a monitoring server.
- **Application Layer:** Real-time dashboards or alert systems for predictive maintenance and environmental supervision.

By integrating FBG-based sensing with IoT communication protocols, the proposed system offers high-resolution, multi-point thermal monitoring with the robustness of optical fiber technology and the connectivity of modern embedded systems. This synergy positions FBG networks as a promising solution for smart city infrastructures, power grid monitoring, pipeline surveillance, and remote safety applications.

V. CONCLUSION

The OptiSystem-based simulation demonstrated that the proposed distributed FBG network maintained high spectral and thermal stability over the temperature range of 10 °C–50 °C.

All five FBGs showed very small wavelength deviations (± 0.001 nm), confirming excellent repeatability, negligible cross-interference, and strong network stability.

These findings suggest that while the system's thermal sensitivity is low, its optical stability makes it ideal for reliable IoT sensing where consistency and low drift are required, such as in environmental monitoring, structural integrity tracking, and industrial process supervision [1], [5], [12].

VI. REFERENCES

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